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When You're There (WYT)

Smartphone Task Reminder Using Indoor Localization

Project Proposal and Requirements Specification by

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ABSTRACT

Due to a lack of a reliable positioning system, indoor location-based services are hindered. Signal blockage and multipath effects also render GPS ineffective in indoor environments. Reliance on smartphone sensors and wireless signals requires common techniques that come with their own limitations. Pedestrian Dead Reckoning (PDR) suffers from cumulative drift errors that degrade over time. Additionally, Wi-Fi positioning based on Received Signal Strength Indication (RSSI) is prone to inaccuracies due to signal fluctuations, multipath fading, and non-line-of-sight (NLOS) conditions.

By developing a sensor fusion framework that integrates data from a smartphone's inertial measurement unit (IMU) sensors and Wi-Fi RSSI measurements, this project intends to combine multiple positioning techniques, leveraging their strengths to create a more robust indoor localization system. The framework consists of a PDR algorithm that uses a device's built-in components, such as the accelerometer, gyroscope, and magnetometer, for relative position tracking. Incorporation of a Wi-Fi positioning module estimates the location by fusing the results from location fingerprinting and trilateration algorithms. Once a position estimate is obtained from the PDR and Wi-Fi modules, they are integrated using a Linear Kalman Filter (LKF), which recursively processes the data stream, producing a more accurate location estimate.

Subject Descriptors: Human-centered computing -> Ubiquitous and mobile computing -> Ubiquitous and mobile computing systems and tools -> Location-based services

Keywords: Indoor localization, Sensor fusion, Wi-Fi positioning, Pedestrian Dead Reckoning (PDR)

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Abbreviations

Abbreviation	Description
ABCL	Adaptive Bayesian Comprehensive Learning
ANN	Artificial Neural Network
AP	Access Point
ASO-LSTM	Aquila Swarm Optimization-Long Short-Term Memory
BLE	Bluetooth Low Energy
CKF	Cubature Kalman Filter
CSI	Channel State Information
DL	Deep Learning
DRL	Deep Reinforcement Learning
EKF	Extended Kalman Filter
FR	Functional Requirement
FTM	Fine-Timing Measurement
GNSS	Global Navigation Satellite System
IMU	Inertial Measurement Unit
IRPLS	Interval Random Parameter Lognormal Shadowing
ISDL	Improved Sequential Deep Learning
IoT	Internet of Things
IPF	Improved Particle Filter
KF	Kalman Filter
LBS	Location-Based Services
LKF	Linear Kalman Filter
MAE	Mean Absolute Error

MEE	Mean Euclidean Error
MEMS	Micro-Electro-Mechanical Systems
NFR	Non-Functional Requirement
NLOS	Non-Line-Of-Sight
OOADM	Object-Oriented Analysis and Design Methodology
PCA	Principal Component Analysis
PDR	Pedestrian Dead Reckoning
PF	Particle Filter
PSO	Particle Swarm Optimization
RSS	Received Signal Strength
RSSI	Received Signal Strength Indication
RTT	Round Trip Time
SLAM	Simultaneous Localization And Mapping
SLE	Step Length Estimation
SMO	Spider Monkey Optimization
UKF	Unscented Kalman Filter
UWB	Ultra-Wideband
VRT	Virtual Reference Tag
WKNN	Weighted K-Nearest Neighbour
xLSTM	Extended Long-Short Term Memory

CHAPTER 01: INTRODUCTION

1.1 Chapter overview

This chapter covers the background and motivation for the development of a smartphone-based indoor localization system. Outlining key challenges and existing systems, this chapter presents the proposed solution and its primary contributions, while providing insight into its conception and expected requirements for execution.

1.2 Problem background

- The increase in availability and usage of mobile smartphones in recent years has caused a demand for mobile services, including indoor location-based services (LBS) *(Zafari et al., 2019)*. People spend the majority of their time indoors, either at home or in buildings where they work, hence the requirement for robust indoor services arises. From warehouse management to hospital patient and equipment monitoring, in larger-scale indoor environments, easy navigation has been proven necessary for higher efficiency *(Mostafa et al., 2025)*.

1.2.1 Inadequacies of Satellite-Based positioning

- While the Global Navigation Satellite System (GNSS) is quite useful outdoors, it has proven to have significant issues with indoor navigation *(Qi et al., 2024)*. The complexity of indoor structures, combined with certain building materials, can block or cause severe signal reflection and multipath effects, thus interfering with the accuracy of location detection, which then renders GNSS unsuitable for indoor navigation, and the need for alternative methods arises *(Poulose et al., 2019)*.

1.2.2 Limitations of exclusively smartphone-based technologies

- To find a resolution for this, many existing fields of research and implementation focus on utilizing standalone smartphone components. The most prominent technologies are PDR and Wi-Fi positioning *(Poulose et al., 2019)*. While these technologies are more effective than GNSS, in isolation, each one suffers from critical flaws that curb their accuracy.

1.2.2.1 Pedestrian Dead Reckoning (PDR) and Cumulative Drift Error

- PDR is an infrastructure-free technique that uses a smartphone's built-in Inertial Measurement Unit to track the user's path. Step detection and step length estimation are then used to identify the user's position relative to a known starting point *(Zhou et al., 2024)*. The primary flaw with PDR is its susceptibility to cumulative drift error caused by low-cost Micro-Electro-Mechanical System (MEMS) sensors that are inherently noisy and lack the precision required for a dependable system *(Wu et al., 2024)*. These erroneous readings are processed and integrated over time, causing dissimilarities between calculated and true trajectories *(Zhou et al., 2024)*.

1.2.2.2 WiFi positioning and signal instability

- WiFi positioning is another widely studied approach, as it is abundant and does not require additional hardware components *(Zafari et al., 2019)*. This is achieved by measuring the Received Signal Strength Indication (RSSI) from nearby Access Points (APs). Unfortunately, this method is compromised due to the instability of RSSI signals *(Poulose et al., 2019)*. RSSI signals are prone to instability and fluctuations caused by;
    - Multipath Fading: signals reflected off various structures and received via multiple paths, causing distortion. *(Naser et al., 2023)*
    - Non-Line-Of-Sight (NLOS) conditions: Physical obstructions such as furniture or walls cause signal attenuation. *(Poulose et al., 2019)*
    - Changes in environment: changes in the layout of the environment, as well as the presence of people, can cause unexpected fluctuations in RSSI signals. *(Zafari et al., 2019)*

1.3 Problem definition

This project addresses the significant inaccuracy and lack of reliability of current indoor localization methods, which prevent the advancement of robust location-aware applications that do not require external components. GNNS exhibits a high degree of accuracy outdoors, but suffers from signal attenuation and interference indoors due to blockage or multipath effects, requiring an alternative technology *(Zhou et al., 2024)*. The two main technologies readily available on smartphones are inertial and wireless signal sensors, but each of them presents its own flaws. Use of wireless signals like Wi-Fi and Bluetooth typically uses RSSI, which suffers from extreme signal instability due to multipath propagation, NLOS conditions, and dynamic environment changes, thereby degrading the accuracy of both the ranging and fingerprinting techniques. Additionally, fingerprinting requires a labor-intensive site survey to create a radio map, which then becomes outdated with environmental changes *(Yu et al., 2020)*.

An alternative to these would be PDR, which uses a phone's built-in IMU, but this is limited by cumulative error or drift *(Poulose et al., 2019)*. Minor inaccuracies in sensor reading accumulate over time, causing the calculated trajectory to diverge significantly from the true path. Furthermore, variation in user behaviour, such as changes in carrying position, can also add to the reading inaccuracies *(Deng et al., 2018)*.

Yet another critical deficiency in many existing systems is the failure to address multi-floor ambiguity. Reliably identifying a user's floor is more vital than the precision of their 2D coordinates, but is affected by the environmental sensitivity of barometers and signal space ambiguity of radio signals *(Mostafa et al., 2025)*.

1.3.1 Problem Statement

- There is a lack of a robust, accurate, and infrastructure-independent indoor localization system based solely on smartphones that can overcome the limitations of PDR and signal interference, especially in complex multi-floor environments.

1.4 Research motivation

This project aims to develop a reminder that uses indoor locations, rather than the standard use of time, to allow users to use their time and energy efficiently. The intended product would allow users to set reminders to complete certain tasks when they are in the vicinity of its respective indoor location. The use of a smartphone's built-in components is intended to reduce the complexity of requirements from the user's end, ensuring the reliability of the reminders. People with memory-related complications, such as alzheimer or even ADHD, could get work done faster by setting a reminder when they are aware of a task that needs to be completed, and simply going about their day until they are in range of the task's location and the application alerts them of it.

1.5 Existing work

Table 1: Existing Work

Citiation	Summary	Contribution	Limitation
(Sun et al., 2025)	Uses WiFi RTT, magnetic field data, and PDR using improved particle filters in a fusion system. Also uses the Cramer-Rao Lower Bound for outlier detection and allows the use of WiFi only when available.	Allows for continuous meter-level positioning even when transitioning into areas without WiFi, enhancing robustness.	Relies on WiFi protocols for RTT measurements, which are not supported by all smartphones and access points, limiting its applicability.
(Lin & Yu, 2024)	Uses a Particle Filter and a Backpropagation neural network for network error reduction when integrating PDR and WiFi indoor mapping constraints.	The use of Particle filters and map constraints improves heading estimation and enhances its accuracy.	Requires a detailed indoor map, and the neural network adds significant computational complexity.
(Zhou et al., 2024)	Fuses PDR with activity semantics, such as turning corners, and map matching to rectify cumulative drift without a known initial position.	Uses a noble method of using topological map features to reduce reliance on the initial known location, correcting PDR errors.	Dependent on an already established indoor map and the accuracy of the activity detection model.
(Yang et al., 2024)	Presents a robust particle filtering method that integrates IMU (PDR), magnetic field, Bluetooth, and a semantic map	The use of a semantic map provides strong environmental context, which prevents trajectory drift into physically impossible locations.	Requires Bluetooth beacons, adding an extra layer of infrastructure and cost. Requires a pre-built semantic map, which is labor-intensive.
(Kim & Shin, 2025)	Proposes the use of a smartphone's	Substantially reduces the	Relies on a pre-built

	magnetometer in combination with a deep learning method to create a magnetic field map for 3D position initialization and drift correction.	complexity of initial position estimation, which is a major challenge for PDR systems.	magnetic field map, requiring a dedicated offline map similar to WiFi fingerprinting.

1.6 Research gap

Existing research shows that a viable approach for smartphone-based indoor localization is the fusion of PDR and WiFi techniques (*Poulose et al., 2019).* Many high-accuracy solutions also depend on tedious pre-deployment surveys to create offline maps *(Kim & Shin, 2025; Lin & Yu, 2024; Zhou et al., 2024)* and identify the initial position. While some address infrastructure dependency *(Kim & Shin, 2025)*, they are susceptible to erroneous data collection caused by varied user movements of smartphones *(Deng et al., 2018)*. The identifiable gap here is the requirement for a lightweight, adaptive fusion framework that balances accuracy, minimizes pre-deployment complexity, while maintaining real-time performance for practical application on commodity hardware.

1.7 Contribution to the body of knowledge

1.7.1 Problem domain contribution

- This project aims to deliver a functional smartphone application that provides valuable indoor location-based reminders without the requirement of additional external hardware or equipment.

1.7.2 Research domain contribution

- This research will contribute a lightweight and robust sensor fusion algorithm that is optimized for real-time localization on everyday smartphones. Using computational filters and methods to handle sensor noise and PDR drift, this work aims to provide a balance between accuracy and ease of development, providing a practical advancement in the field of indoor localization.

1.8 Research challenges

- **Accuracy vs Efficiency**: Achieving high localization accuracy often requires frequent sensor sampling and computation, requiring a lot of energy from the device *(Zafari et al., 2019)*.
- **Signal instability**: On its own, WiFi RSSI is highly susceptible to fluctuations caused by multipath propagation, obstacles, and environmental changes, which makes it unreliable *(Poulose et al., 2019)*.
- **Sensor drift**: IMU sensors in smartphones are prone to noise and sensitive to erroneous stimuli, leading to cumulative drift errors in PDR that can render long-term tracking inaccurate *(Poulose et al., 2019)*.
- **User and device heterogeneity**: The effectiveness of the system relies on a smartphone's ability to measure and compute specific readings from sensors, which may be limited to certain models and user behaviors *(Zhou et al., 2024)*.

- **Real-time performance**: Time of computation is directly proportional to its complexity, so achieving a balance in high computation while being simple enough to reduce lag is quite challenging *(Mostafa et al., 2025).*

1.9 Research questions

- How can data from smartphone IMU sensors and WiFi RSSI be effectively fused to achieve high localization accuracy while not requiring external hardware?
- How can PDR accuracy be optimized against drift and erroneous data collection from unpredictable external factors, including changes in carrying position?
- What filtering algorithm provides the best trade-off between localization accuracy, computational efficiency, and power consumption for the best real-time application for a standard smartphone?

1.10 Research aim and objectives

1.10.1 Research aim

- This research aims to design, develop, and evaluate a robust and efficient indoor localization application for smartphones that provides location-based reminders by fusing data from built-in IMU sensors and Wi-Fi signals.

1.10.2 Research objectives

Table 2: Research Objectives

Objectives	Research objective	Description	LOs Mapped	RQ Mapped
Problem Identification	To identify the research gap in the area of indoor localization	Look into existing research in the field of indoor localization and find a critical gap with existing research	LO2, LO4	RQ1,RQ2
Literature Review	To find indoor localization techniques in existing literature	Conduct a comprehensive literature survey on smartphone-based indoor localization techniques to find the most appropriate methods and technologies currently studied.	LO1, LO4	RQ2,RQ3
Requirement Elicitation	To identify the functional and non-functional requirements for a smartphone-based indoor location reminder	Identify functional(location setting, reminder triggering, etc) and non-functional(accuracy, real-time performance, battery consumption, etc) requirements	LO3	RQ1
Legality and ethics	To address legal, social, and ethical implications related to indoor	Location data pertains to users' private information, and as such, identifying and avoiding	LO6	RQ3

	localization.	infringement on this is key.		
System design	To design the software architecture for a multi-sensor fusion system.	Identify the components, tools, and techniques required and design the architecture for the final application.	LO1,LO5	RQ2,RQ3
Objectives	**Research objective**	**Description**	**LOs Mapped**	**RQ Mapped**
Implementation	To implement a PDR module that uses smartphone IMU sensors to collect data.	Create a module that performs step detection, step length estimation, and heading direction by processing raw data from smartphone IMU sensors.	LO1, LO5, LO7	RQ1,RQ3
	To implement a WiFi positioning module.	Create a module that performs WiFi positioning using trilateration and/or fingerprinting techniques based on RSSI measurements.	LO1, LO5, LO7	RQ1,RQ2
	To develop a lightweight fusion algorithm.	Develop a fusion algorithm, specifically a Linear Kalman Filter, to integrate the position estimates from the PDR and WiFi module in real-time.	LO1, LO5, LO7	RQ2,RQ3
	To develop a prototype Android application with a functional user interface.	Create the prototype for the application such that it allows the user to set, view, and receive reminders triggered by the estimated position.	LO7	RQ2,RQ3
Testing & Evaluation	To evaluate location accuracy and real-time performance	Understand the overall accuracy and performance of the positioning system by comparing its position estimates against pre-recorded true data within a test environment.	LO1, LO8	RQ3
Documentation and Defense	To produce detailed documents containing all the critical information from the project life-cycle, with corresponding defense for each.	The documentation must evaluate the project plan, the new skills acquired, and the successful demonstration of the final application.	LO2, LO5, LO8, LO9	R1,RQ2, RQ3

1.11 Project scope

1.11.1 In scope

- PDR core implementation: Development and implementation of a PRD algorithm using a smartphone's built-in IMU sensors, providing continuous monitoring of relative position.
- Pose-invariant tracking: Incorporate techniques required to minimize sensor data collection errors caused by changes in carrying position, irrelevant of actual changes in location.
- Absolute position sources: Implementation of the absolute position localization method to correct the inherent accumulation of drift, commonly seen with PDR.
- Opportunistic use of WiFi: Utilise WiFi RSSI when wifi is available to enhance accuracy, while not being entirely dependent on it.
- Magnetic field mapping: Make use of the smartphone's magnetometer to leverage unique indoor magnetic field patterns to be used in combination with other IMU sensors to further drift correction or map matching.
- Application layer: Development of a prototype smartphone application that acts as a user interface for this localization and reminder system.

1.11.2 Out scope

- Device heterogeneity: The project will be tested and limited to the software specifications of a maximum of two chosen smartphone devices with similar specifications.
- Storage optimization: Fine-tuning of the application to carry out all functionality while maintaining extremely low storage capacity is out of scope for the prototype.
- High-speed real-time computation: While efficiency and speed are core considerations, it is out of scope for a proof of concept that primarily aims to test and prove the feasibility of such a system.
- Indoor-Outdoor transition: The system focuses only on indoor localization and avoids anything regarding outdoor transition and localization.
- Multi-directional movement with uni-directional facing: The system considers the direction being faced in addition to movement readings. Movement in directions contradicting the direction being faced is out of scope.
- Energy optimization: As a proof of concept, the system will keep energy consumption to a manageable degree rather than full optimization to allow smartphone components to be used to the maximum capacity.

1.12 Hardware/Software requirements

Table 3: Hardware/Software Requirements

Category	Requirement	Justification
Hardware	Commodity Android Smartphone (e.g., Google Pixel, Samsung Galaxy Series) with IMU, magnetometer, barometer, and Wi-Fi capabilities.	To ensure the solution applies to widely available consumer devices while also having the most vital components required from a smartphone.
	Development PC with a dedicated GPU (e.g., NVIDIA RTX 30-series or higher, >8GB VRAM).	Required for efficient training of the deep learning models.
Software	Android studio	For developing and deploying the prototype application.
	Python 3.8+ with libraries: PyTorch or TensorFlow, Scikit-learn, NumPy, Pandas.	For data processing, model development, training, and evaluation.
	Google Docs	For documentation
	Google Sheets	For data organization and clear planning.

1.13 Chapter summary

This chapter establishes the context and significance of developing a robust smartphone-based indoor localization system. It outlines the weaknesses of standalone PDR and WiFi methods, identifies the research gap for an effective fusion system, and defines the project's aim, objectives, scope, and requirements.

CHAPTER 02: LITERATURE REVIEW

2.1 Chapter overview

This chapter covers an in-depth review of existing literature on indoor localization, breaking down their contribution and shortcomings to shed light on the path taken to reach the proposed solution. It establishes the technical challenges inherent in indoor positioning, analyzes the limitations of minimal method-dependent systems, and synthesizes state-of-the-art sensor fusion frameworks, particularly those leveraging Deep Learning (DL) and environmental constraints to maintain high-level accuracy.

2.2 Problem domain

Indoor localization is driven by the necessity to provide accurate and reliable real-time location-based services in environments where traditional GNSS, such as GPS, is ineffective. Developments in this domain have witnessed a surge in interest due to the widespread availability of smartphones and other smart devices, which serve as platforms for the boom of Internet of Things (IoT) technologies *(Zafari et al., 2019)*.

People spend a significant amount of time indoors, leading to an increase in demand for indoor quality-of-life improvements through increasingly robust real-time systems for scenarios such as patient monitoring, asset management in warehouses, mall navigation, and emergency evacuation *(Qi et al., 2024),* all of which require high-precision positioning to be considered a viable tool to be used.

2.2.1 Core challenges of indoor localization

Indoor environments are often unique, presenting their own significant challenges in comparison to traditional localization methods.

2.2.1.1 GNSS Ineffectiveness

- Satellite signals suffer severe attenuation, degradation, blockage, and multipath effects when attempting to penetrate building structural supports like walls and roofs. This renders GNSS inaccurate and ineffective indoors *(Poulose et al., 2019)*.

2.2.1.2 Harsh radio environments

- Indoor spaces are dynamic and cluttered, leading to frequent multipath effects, reflection, diffraction, and scattering of wireless signals further hindered in NLOS conditions *(Leitch et al., 2023)*.

2.2.1.3 Lack of standardization and infrastructure cost

- Unlike outdoor localization, there is no set standard or widely accepted technology for indoor localization *(Naser et al., 2023)*. Currently studied, high-accuracy technologies such as UWB require specific dedicated infrastructure, increasing system cost and complexity *(Zafari et al., 2019)*.

2.2.2 Limitations of single-technology systems

To address the infrastructure constraint, the focus shifts to technologies that leverage existing hardware, primarily the sensors embedded in commercial smartphones *(Leitch et al., 2023)*. However, relying solely on a single technology presents its own challenges.

2.2.2.1 Pedestrian Dead Reckoning (PDR)

- Based on smartphone IMU sensors, it is cost-effective and infrastructure-independent. However, PDR suffers from cumulative drift errors due to the low precision of consumer-grade MEMS sensors *(Poulose et al., 2019)*. Its accuracy is also hindered by the collection and computation of complex handheld motion patterns, not related to changes in their indoor position *(Yu et al., 2023)*.

2.2.2.2 WiFi positioning (RSSI Fingerprinting)

- The abundance of existing Access Points (APs) makes WiFi-based positioning a major point of focus for localization. Fingerprinting involves the mapping of RSSI values at known reference points to create radio maps *(Zhang et al., 2024)*. Unfortunately, RSSI is highly susceptible to fluctuation and requires tedious, time-consuming site surveys and frequent updates of the radio map, making maintenance costly *(Naser et al., 2023)*.

2.2.2.3 Magnetic field localization

- Magnetic signatures of indoor environments are unique and can provide absolute location fixes *(Qi et al., 2024)*. However, the performance is affected by the magnetic sensor distortion caused by steel structural components within buildings and interference from other electronic devices *(Naser et al., 2023)*.

2.3 Existing work

The search for accurate, continuous, and infrastructure-free indoor localization uses current state-of-the-art that focuses on overcoming the limitations of single-sensor systems, namely the inherent drift in PDR and the instability of WiFi signal-based positioning through means of sensor fusion techniques, often incorporating Machine Learning (ML) and Deep Learning (DL) *(Poulose et al., 2019)*.

2.3.1 Standalone localization techniques

Before fusion, individual techniques were explored, each with its own trade-offs.

2.3.1.1 WiFi RSSI fingerprinting

- As described in “2.2.2.2 WiFi positioning”, this is the most popular approach due to the abundance of APs. However, the localization accuracy depends on the amount of fingerprint data generated, and typical accuracy hovers around 1.5-2m due to signal fluctuations and multipath effects *(Poulose et al., 2019)*. It also requires a labour-intensive offline location survey to generate and maintain the radio map.

2.3.1.2 WiFi Fine-Timing Measurement (FTM) / RTT

- This is capable of achieving sub-meter accuracy, with one study reporting 0.39m accuracy when tightly coupled with PDR and RSS *(Wu et al., 2024)*. However, it requires specific device support and can exhibit poor performance in NLOS conditions and high traffic scenarios *(Leitch et al., 2023)*.

2.3.1.3 Ultra-Wideband (UWB)

- Offers high accuracy with errors as low as 0.03m to 0.3m in some cases, and exhibits superior object penetration and minimal interference with other signals *(Poulose et al., 2019)*. The only drawback is that it requires higher implementation and deployment costs *(Leitch et al., 2023)*.

2.3.1.4 Inertial sensor-based methods (PDR)

- Uses built-in smartphone sensors for infrastructure-free, continuous tracking, but suffers from accumulated positioning error (drift) that increases with walking distance due to the inherent low precision of consumer-grade MEMS sensors. It is also greatly affected by changes in carrying positions being read and incorporated into the computation.

2.3.1.5 Geomagnetic localization

- As mentioned previously, it can be used for fingerprinting, providing an infrastructure-free correction node (CN) to mitigate PDR drift. However, it also suffers from fluctuations affecting system performance.

2.3.2 Fusion frameworks and advanced algorithms

To overcome the weaknesses of single sources, multi-source sensor fusion has become another point of research focus. Fusion methods are commonly categorized as filtering-based or learning-based approaches.

2.3.2.1 Filtering-based fusion (Analytics-based)

Traditional analytic approaches integrate state prediction (PDR) with observation updates from sensors and wireless signals *(Poulose et al., 2019)*.

- Kalman Filters (KF):
  - These are effective for linear systems *(Liu et al., 2025)*. The Extended Kalman Filter (EKF) and Unscented Kalman Filter (UKF) are approximate nonlinear models, but EKF struggles with severe nonlinearity and non-Gaussian noise. An EKF can be used to fuse WiFi and inertial sensors for localization, while UKFs have been proposed for integration of WiFi localization and PDR in 3D environments. KF-based systems typically rely on accurate system models and noise assumptions *(Poulose et al., 2019)*.

- Particle Filters (PF):

- These are highly utilized in fusion systems, particularly in conjunction with WiFi and IMU data. The PF is a recursive filter based on the sequential Monte Carlo framework *(Liu et al., 2025)*. It is considered a traditional analytical approach that integrates state prediction with observation updates, thereby approximating the true posterior distribution of the system data *(Poulose et al., 2019)*. The position and heading are typically estimated by computing a weighted average across a set of particles, where each particle represents a state for the pedestrian *(Yang et al., 2024)*.

2.3.3 Literature Survey

Table 4: Literature Survey

Citation	Summary	Limitations
(Fahama et al., 2025)	Uses RSSI fingerprints collected from multiple wireless technologies to train supervised ML models to balance accuracy, precision, installation time, and labor.	Supervised Machine Learning techniques generally require high computational complexity during the training phases. Relying solely on a trained ML model requires time-intensive steps, including an extensive measurement and data collection offline phase.
(Kim & Shin, 2025)	Proposes using a Deep Learning model trained on magnetic field data for accurate 3D location initialization using only the user's first 10 steps. Confirms the use of magnetic correction nodes to mitigate PDR drift.	The solution assumes that a position transition and a user turn cannot co-occur. It only investigates forward walking and does not consider other walking modes.
(Liu et al., 2025)	This is a review of work regarding multi-source sensor fusion methods, classifying and analysing smartphone-based single-sensor and multi-source fusion localization techniques.	It can be improved by the inclusion of Bluetooth and/or PDR to enhance its accuracy.
(Mao et al., 2025)	This paper proposes the utilization of an Improved Sequential Deep Learning (ISDL) model combined with a robust WKNN fingerprint-based indoor positioning in multi-floor environments using a tree-based classification method to filter training samples.	Use of a Deep Learning based approach is inefficient on larger scales, as a new model must be created for each new environment, requiring extensive training and data collection each time.
(Mostafa et al., 2025)	Provides a framework for highly portable, software-based floor detection by localizing users within a virtual building composed of a constant window of floors and relative AP locations.	The system relies on external WiFi APs and RSS measurements for floor prediction input.

(Sun et al., 2025)	Proposes the integration of WiFi RTT, magnetic field data, and PDR using an IPF optimized with map-aided resampling and particle swarm reconstruction. It ensures continuous localization by switching between WiFi/magnetic/PDR fusion and magnetic/PDR fusion when WiFi is unavailable.	WiFi RTT ranging requires specific device support. Future work also discusses the integration of WiFi FTM fingerprinting to enhance the accuracy.
(Lin & Yu, 2024)	Proposes improvements to PDR and Weighted k-nearest neighbour (WKNN) positioning algorithms by using map constraints (such as corners). This highlights the value of map data and refined PDR methods in improving the system's accuracy.	The proposed solution requires integration with WiFi localization and involves the offline establishment of corner coordinates and constrained angles in a database.
(Mathavan & Sivaranjani, 2024)	Proposes an RFID localization and tracking method that employs Virtual Reference Tags (VRTs) allocated optimally using Spider Monkey Optimization (SMO), Artificial Neural Network (ANN), and Aquila Swarm Optimization-Long Short-Term Memory (ASO-LSTM) for tracking.	As an RFID-based system, it faces challenges related to signal blocking and NLOS conditions. Although RFID is suitable for large-scale environments, it requires the deployment of maintenance of RFID infrastructure. This also makes this solution not entirely infrastructure-independent.
(Qi et al., 2024)	Proposes the use of EKF/UKF fusion structures, particularly when combined with WiFi FTM and multiple sensors, to provide a robust analytic technique for merging continuous PDR movement with periodic range observations.	This specializes in WiFi FTM anchors/APs, which may not be readily available or infrastructure-independent.
(Wu et al., 2024)	Demonstrates sub-meter positioning capability using an advanced filtering technique (IPF-UKE) to fuse IMU data with surface electromyography (SEMGS) sensors, indicating high accuracy and stability for PDR methods.	Most comparable PDR studies require calibration of the magnetic heading before use, which is difficult to achieve in real-time. The proposed PDR method only considers walking forward.
(Yang et al., 2024)	Uses a PF framework to integrate IMU, magnetic sensors, and semantic map constraints, thereby enhancing positioning robustness by assigning probability constraints based on passable areas.	Requires external infrastructure (BLE base stations) for localization constraints and initial particle position distribution. Relies on pre-modeling and scene evaluation using SLAM equipment.
(Zhang et al., 2024)	This IRPLS-ABCL localization model uses an interval random parameter longnorm	Although improved in accuracy, the enhanced algorithmic complexity

	shadowing model to automatically enhance the fingerprint radio map and employs a self-adaptive Bayesian comprehensive learning PSO algorithm for optimization, outputting an optimal location area instead of precise coordinates.	unavoidably extends the algorithm's execution time.
(Zhou et al., 2024)	Confirms the viability of correcting the cumulative drift error problem of PDR by using detected activities as virtual landmarks for map matching.	Requires accurate mapping of where each of these activities occurs (The landmark locations).
(Csik et al., 2023)	This method proposes the fusion of fingerprints collected using different radiocommunication technologies using machine learning algorithms to exploit the distinct behavior of these technologies indoors and to improve localization efficiency.	Range-based methods are limited by requiring the target position to be within the coverage range of the reference nodes. Time measurement components require precise clock synchronization to avoid performance degradation.
(Leitch et al., 2023)	Deep Reinforcement Learning (DRL) algorithms can significantly improve localization accuracy, achieving a 48.19% improvement over traditional methods in BLE RSS fingerprinting experiments	The high-accuracy DRL method requires external beacon infrastructure as it relies on BLE RSS fingerprinting.
(Naser et al., 2023)	This covers and summarizes smartphone-based indoor localization, reviewing technologies, methods, motivations, and challenges reported in literature from 2015 to 2022.	Highlights the need for consideration of various phone poses and motion states, as well as addressing device heterogeneity, which refers to sensor value gaps between different smartphone models.
(Yu et al., 2023)	Supports the possibility of using built-in sensors for the 3D localization of positioning when fused with WiFi FTM, confirming that FTM data can be analyzed and processed for localization.	Requires external WiFi FTM infrastructure.
(Poulose et al., 2019)	Confirms the viability of fusing multiple technologies, including magnetic matching and PDR, to enhance indoor navigation and accuracy	Incorporates WiFi as a necessity, which often requires external AP infrastructure and extensive site surveying.
(Zafari et al., 2019)	Crowdsourcing offers a solution to the traditional labor-intensive site survey required for localization, aiming for accurate localization with zero start-up cost.	This system relies a lot on the presence of WiFi signals and the integrity/accuracy of crowdsourced map updates.
(Deng et al., 2018)	Proposes a novel heading estimation system	While it takes hand movement errors

	designed to adapt to the unconstrained use of smartphones, addressing issues related to varying device carrying positions and orientations.	into account, it is still susceptible to cumulative errors with PDR drift.

2.4 Dataset selection

The selection of appropriate datasets is vital for the development and evaluation of indoor localization and tracking models. The supporting datasets typically comprise various forms of sensor data and radio frequency measurements collected in controlled or real-world environments.

2.4.1 WiFi and radio frequency fingerprinting datasets

This aspect relies heavily on large-scale, pre-collected RSSI or related radio data linked to known coordinates.

2.4.1.1 Dataset: UJIIndoorLoc *(Mao et al., 2025)*

- UJIIndoorLoc Dataset is a publicly available WiFi fingerprinting database spanning over 108,000m^2 across three to five floors of buildings on the University Jaume I (UJI) campus in Spain. It serves as a benchmark for the evaluation of positioning accuracy in multi-floor environments.
- The dataset includes a training set, comprising 19,937 records, and a validation set, comprising 1,111 records, where each entry contains real-world coordinates in the form of longitude, latitude, and floor, and 520 RSSI values from all detected APs.
- The dataset is robust for assessing generalization, as nine out of eleven devices used in the validation set are distinct from those in the training set, and validation data were collected three months post-training.

2.4.1.2 Dataset: ZigBee RSSI *(Fahama et al., 2025)*

- A specific dataset used for supervised machine learning models was collected in an empty small meeting room (33m^2) using ZigBee wireless technology. This test bench used three APs and featured 49 fingerprint reference points spaced 0.5m apart.

2.5 Benchmarking and evaluation

2.5.1 Benchmarking criteria and qualitative analysis

Table 5: Benchmarking criteria

Metric	WiFi (General)	UWB	BLE	IMU (PDR)
Accuracy	0.09-7.60m	0.03-0.30m	0.10-4.92m	0.55-5.64m
Robustness	Medium	Good	Poor	Good
Scalability	Good	Medium	Good	Excellent

Computational Complexity	Low-Medium	Medium	Low	Low (Device dependent)
Cost	Low	High	Low	Low (Device dependent)
Typical coverage	30-50m	30-50m	Covers a small area	N/A

Insights;

- Accuracy Goal: For indoor localization, the goal is often to achieve an average estimation error of less than 10cm. Only UWB and specific WiFi techniques (like CSI fingerprinting) are capable of achieving this level of accuracy alone.
- Cost and Infrastructure: WiFi localization is highly attractive due to the ubiquity and low cost of its hardware. PDR (IMU-based) is generally the easiest and cheapest to implement as it relies only on the target device. Conversely, UWB hardware, while capable of the highest accuracy, is relatively niche, and installation drives up the system cost.
- Fingerprinting vs Ranging/Trilateration: Fingerprinting performs poorly across cost, complexity, and time metrics, primarily due to the protracted period of site surveying required to build and maintain the radio map. Range-based methods like WiFi RTT are newer and can achieve sub-meter accuracy.

2.5.2 Quantitative benchmarking of localization systems

Comparison of systems often performed using metrics that quantify positional error (e.g., Mean error, RMSE) and temporal performance (e.g., Test Time and Reliability).

1. Benchmarking Fusion Systems (Accuracy and reliability)

Table 6: Quantitative analysis (Fusion systems)

System/Approach	Fusion Components	Key Accuracy Metrics	Value	Comparison Notes
Proposed IPF System	WiFi RTT/Magnetic/PDR	Mean Accuracy (WiFi available)	0.64m	Significantly outperforms individual component methods.
Proposed IPF System	Magnetic/PDR	Mean Accuracy (WiFi unavailable)	0.91m	Maintains meter-level continuity even without WiFi signals.
PDR + WiFi fingerprint (Conventional)	PDR + WiFi Fingerprint	Average Error	1.5-2m	Worse than the proposed sensor fusion framework
Hybrid WiFi/UWB/IMU	CC RSSI + UWB RSSI + UWB TOF	Mean Absolute Error (MAE)	30.23cm	Outperforms most accurate single technology by nearly 10%.
Deep Learning xLSTM	UWB/IMU (Residual Fusion)	Mean Euclidean Error	0.18m	Lowest MSE/MEE compared to other deep learning and

		(MEE)		filter-based fusion strategies (e.g., Attenuation: 0.23m)
Traditional Filters	UWB/IMU (CKF)	Accuracy (≤ 0.25m range)	44.4%-47.9%	Proposed xLSTM improved accuracy in this range by 60.1% to 64.8% over CKF.

2. Benchmarking Deep Learning/Machine Learning Models (Accuracy vs. Time)

Table 7: Quantitative analysis (DL/ML models)

Technique	**Mean Error (m)**	**Standard Deviation (m)**	**Mean Time (ms)**	**Tracking Accuracy (%)**
ANN+ASO-LSTM (RFID)	0.973	0.509	28.2	98.99 ± 0.5
CNN (RFID)	1.130	0.537	27.4	N/A
DNN (RFID)	1.224	0.563	27.4	N/A
LANDMARC (RFID)	1.599	1.001	1.3	N/A
WKNN (RFID)	1.626	1.014	1.3	N/A
CF (ZigBee SML)	1.2164 (MSE)	6001.1 (Variance)	N/A	N/A
FF (ZigBee SML)	1.3324 (MSE)	N/A	N/A	N/A

- Trade-offs in ML Benchmarking: Deep learning models (ANN+ASO-LSTM, CNN, DNN) typically offer significantly lower mean errors and higher tracking accuracy compared to simpler methods like LANDMARC and WKNN. However, simple methods like LANDMARC and WKNN demonstrate substantially lower computational complexity, achieving mean execution times of 1.3 ms, versus the deep learning methods, which range from 17.6 ms to 28.2 ms.

3. Benchmarking Efficiency and Latency

- Processing time and latency are crucial metrics, especially for real-time applications.
    - The proposed Wi-Fi RTT/magnetic/PDR fusion system completed every fusion calculation within 0.1s.
    - The xLSTM residual fusion method achieved a very low inference latency of approximately 17–18 milliseconds.
    - The IRPLS-ABCL localization model, while highly accurate (1.89 m mean error), has an enhanced algorithmic complexity that unavoidably extends the algorithm's execution time compared to other algorithms, though it still meets real-time requirements in practical applications.

- For BLE proximity detection, a Bluetooth Location Network (BLN) system achieved a response time of about 11 seconds, making it non-real-time. The use of iBeacons, due to inherent framework limitations, typically results in significant delay.

2.6 Chapter summary

This chapter presents an in-depth literature review that emphasizes the necessity of indoor localization, as GNSS is ineffective indoors due to signal attenuation and multipath effects caused by building structures. The domain lacks standardization, and high-accuracy systems, like UWB, are costly due to additional infrastructure requirements. On the other hand, low-infrastructure systems rely on singular technologies and suffer major limitations. PDR suffers from cumulative drift errors, and WiFi RSSI fingerprinting is highly susceptible to fluctuations, requiring tedious site surveys. Consequently, current research focuses on multi-source sensor fusion, incorporating machine learning and deep learning, as well as traditional analytical methods such as EKF and PF. These filter-based frameworks integrate the continuous PDR state prediction with periodic observation updates from wireless signals to enhance robustness. The performance of these models is then evaluated using large-scale benchmarks such as the UJIIndoorLoc Dataset, a publicly available WiFi fingerprinting database.

CHAPTER 03: METHODOLOGY

3.1 Chapter overview

This chapter outlines the methodology for a robust indoor localization system via multi-source sensor fusion, designed to overcome single-sensor limitations. It details methods to improve PDR, including the use of magnetic constraints and integrating absolute location data from WiFi/BLE/UWB sources. Core fusion techniques reviewed include advanced filtering, such as IPF and deep temporal feature learning such as Extended Long-Short Term Memory (xLSTM), culminating in a discussion of experimental performance metrics.

3.2 Research Methodology (Saunders' Research Onion in Table Form)

Table 8: Research Methodologies

Layer	Choice	Justification
Philosophy	**Positivism**	The project concerns the evaluation of a hypothesis regarding computational models for indoor localization accuracy, which requires quantitative measurable outputs to validate system performance against the specified constraints.
Approach	**Deductive**	The research starts with existing theories and models for indoor localization (PDR, fusion frameworks, or DL architectures) and applies them to a novel context to determine their feasibility and accuracy, thus testing existing theory.
Methodological Choice	**Multi Method**	**Requirement Elicitation (Intrusive Procedure):** The study initially uses an (Online) Survey/Questionnaire, which is an intrusive procedure, to systematically collect feedback from potential end-users regarding the functional and non-functional requirements (FRs/NFRs) of the application. **Technical Validation and Experimentation:** The core method for technical validation is Experimentation. The study focuses on infrastructure-free localization using smartphone IMU sensors, integrated via Pedestrian Dead Reckoning (PDR).
Strategy	**Experimental, Case Study & Action Research**	**Experimental Design:** Necessary to quantitatively measure the application's localization performance and error in a controlled environment. **Action Research/Case Study:** To iteratively develop the localization system and reminder functionality, gather practical insights, and incorporate feedback directly from users testing the application in a real indoor environment.

Time Horizon	**Longitudinal Study**	Training models, particularly fingerprinting models, requires collecting data at discrete points. However, evaluating the actual application requires monitoring continuous movement over time (**longitudinal/time-series data**) to measure accumulated error (drift) and assess long-term stability, which is essential for PDR-based systems.
Data collection	**Primary Data** and **Archival Research**	**Primary Data**: Necessary to collect real-time sensor measurements using the actual developed application on the smartphone device in the targeted environment to build and test the localization model/fingerprint maps. **Archival Data**: Existing public localization datasets may be used for initial model training, comparison, or validation of ML/DL approaches.

Hypothesis: An infrastructure-independent localization framework utilizing fused smartphone IMU and barometer sensors can achieve continuous and accurate indoor positioning (including floor detection), maintaining high stability and resolving cumulative drift without requiring additional external hardware.

3.3 Development Methodology

The foundation of any complex system development, such as an indoor localization application, requires a clear methodology involving phases like Requirement Elicitation (e.g., Interviews, Document Analysis) and Design methodology. Successful project execution further relies on a robust Solution Methodology, including steps like Feature Selection and Engineering, Model Training, and Testing.

The choice of development methodology for this specific application is an ***Iterative*** approach, implemented through an Experimental Strategy. This is because Iterative methodology is rooted in the high degree of technical uncertainty, the essential requirement for multi-source sensor fusion, and the crucial need for continuous performance tuning inherent in developing a robust, infrastructure-independent indoor localization application on a smartphone.

The application is intended to run entirely on the resource-constrained smartphone, making efficiency as important as accuracy. The iterative approach facilitated the necessary cycles of data collection and algorithm refinement.

3.3.1 Requirement Elicitation

Various methodologies were used to elicit system requirements. Focusing heavily on analyzing existing technical solutions and validating application feasibility while considering the constraints of infrastructure-independent localization.

- **Document Analysis**: Used to define the technical scope and limitations of an infrastructure-independent system, specifically by studying existing literature on sensor-based indoor positioning:

- **PDR Foundation**: The primary method for infrastructure-independent localization is PDR, which uses the smartphone's built-in IMU sensors, including the accelerometer, gyroscope, and magnetometer. Document Analysis is necessary to define the algorithm required for PDR, such as step detection, step length estimation, and heading determination.
- **Addressing error accumulation**: Analyzing existing literature is vital to define requirements for managing PDR's crucial limitation, which is accumulated error drift over time. This points to the need for sensor fusion techniques like Kalman Filters (KF) or Particle Filters (PF) to integrate data from the IMU with other reference sources such as geomagnetic field data.
- **Magnetic Field Utilization**: Reviewing research on using the smartphone's magnetometer is necessary, as magnetic field patterns can be used for location estimation and drift correction without external infrastructure. Magnetic field-based localization relies on the fact that building structures distort the Earth's magnetic field, creating unique, location-specific patterns.
- **Multi-floor capability**: If multi-floor operation is a requirement, the analysis focuses on utilizing the smartphone's barometer to detect altitude changes when a user ascends/descends stairs or uses elevators, thereby solving the robust initial floor detection problem.

- **Observation**: This is essential for capturing real-world variables that directly impact the input data quality from the sensors:
  - **Phone Pose and Movement**: Requirements must account for the highly variable placement of the smartphone, as the sensor readings differ significantly if the user is texting (handheld), swinging the device, or carrying it in a pocket or bag. Observation helps quantify how these varying phone poses affect the raw accelerometer and gyroscope data, which is necessary to define requirements for PDR algorithms that must function reliably across multiple modes.
  - **Walking patterns and context**: Observing users engaging in different activities (normal walking, standing, turning, running) is key to formulating robust feature extraction requirements for activity detection. Identifying turns or stops is crucial for PDR refinement, as these instances can be used as constraints to limit error accumulation.

- **Brainstorming and Self-Evaluation**: This helps establish core constraints vital for a practical smartphone application, particularly defining the Non-Functional Requirements (NFR).
  - **Energy Efficiency (NFR)**: Given that continuous reading of IMU sensors can significantly drain the user device battery, brainstorming requirements focuses on optimizing sensor sampling rates and implementing power-saving strategies to ensure system efficiency.
  - **Computational Cost (NFR)**: Self-evaluation of algorithms is necessary to choose methods, such as lightweight PDR algorithms or filter-based approaches that can run on a standard smartphone processor without excessive latency, ensuring real-time responsiveness for the reminder function.
  - **Localization Precision (FR/NFR)**: Discussion centers on defining the acceptable level of accuracy. Since absolute positioning requires extensive site surveys and maintenance, the application must rely on relative position or robust location area identification (proximity

detection using existing WiFi/BLE RSSI, where available) to provide realistic localization estimates.

3.3.2 Design Methodology: Object-Oriented Analysis and Design Methodology (OOADM)

The OOADM approach is selected because the system requires the seamless integration of various complex, interacting components (e.g., sensor data processing, PDR, sensor fusion filters, and application logic like reminder settings). OOADM supports modularity, reusability, and efficient management of these components, making it ideal for a sophisticated, real-time mobile application.

The design methodology will focus on defining robust, independent objects to handle the challenges specific to infrastructure-independent indoor positioning, namely sensor noise, accumulated drift error, and variable carrying positions.

3.3.2.1 Object-Oriented Analysis (OOA)

Object-Oriented Analysis focuses on identifying the necessary components (Classes and objects) and their requirements based on the technical constraints below:

- **Sensor Data Objects**: These objects manage the continuous stream of raw data from the embedded sensors, including the IMU and barometer.
- **Activity/Pose Recognition Object**: Given that the sensor readings are significantly influenced by the user's activity and device placement, this object must use DL or ML techniques to classify the current phone pose. This recognition is crucial for adapting PDR parameters accordingly.
- **Map Constraint Object**: This object establishes the permissible movement boundaries using a topological map composed of points and lines, typically excluding walls and obstacles.

3.3.2.2 Object-Oriented Design (OOD)

The OOD phase dictates the architecture and specific algorithms used to achieve robust #D localization using only the smartphone:

- **Core Localization Architecture**: The system utilizes a Multi-Source Sensor Fusion Framework to overcome the inherent limitations of individual sensors, particularly the drift associated with PDR. The fusion framework consists of two main processing paths:
    - **Relative Positioning (PDR Module)**: This module calculates the relative change in position for each step using IMU data.
        - **Step Detection and Step Length Estimation (SLE)**: Algorithms must be implemented for step detection (e.g., peak detection) and SLE (adaptive models).
        - **Heading Estimation**: Accurate heading is crucial but challenged by magnetic interference and phone orientation. Heading estimation methods must be employed. Robust heading estimation systems adapt to unconstrained use by detecting and classifying motion states (e.g., hand movements, user turns).

- **Absolute Correction and Initialization (Magnetic/Barometric Modules)**: These modules provide periodic, absolute location references to correct PDR drift.
    - **3D Location Initialization**: The magnetometer measures the Earth's magnetic field, which is distorted by building structures, creating unique floor-specific patterns. Magnetic field map-based location calibration (magnetic fingerprinting) using machine learning algorithms, such as k-Nearest Neighbor, is designed to determine the initial 2D position and floor level.
    - **Floor Transition Detection**: The barometer is used to detect altitude changes, enabling accurate floor detection when using stairs or elevators.

- **Data Fusion Engine**: A Particle Filter (PF) is designed as the primary fusion algorithm due to its effectiveness in tightly integrating non-linear PDR measurements with noisy absolute position measurements (magnetic/barometric) and incorporating map constraints.
    - **Functionality**: The PF maintains particle swarms representing possible user locations. It propagates particles using PDR estimates during the prediction phase and weights them based on matching with the magnetic field map and constraints during the update phase.
    - **Enhancements**: The PF design should include map-aided resampling and particle swarm reconstruction strategies to reduce error accumulation and prevent particles from tracking infeasible routes (e.g., walking through walls).

3.3.3 Programming Paradigm

The **Object-Oriented Programming (OOP)** approach was selected to facilitate modular design and manage the complexity of multi-source sensor fusion algorithms (such as Pedestrian Dead Reckoning and IMU-based localization) necessary for robust, infrastructure-independent localization running entirely on the smartphone.

3.3.4 Programming Paradigm

Experimental Validation will be used to ensure the accuracy and stability of the infrastructure-independent localization system, counteracting the cumulative error inherent to smartphone IMU sensors and the variability introduced by unconstrained usage, by performing extensive pedestrian positioning experiments across predefined paths, evaluating performance across diverse smartphone usage modes (e.g., texting, swinging, pocket, bag) and specific motion states (e.g., hand movements, user turns, position transitions) against ground truth data.

3.3.5 Solution Methodology (Expanded Subsections)

1. **Dataset Collection**: For an infrastructure-free localization system, the dataset collection focuses on gathering raw sensor measurements and establishing environmental maps for calibration and initialization.

- **Sensor Data Acquisition:** Raw data must be collected from the smartphone's built-in sensors, including the accelerometer, gyroscope, magnetometer, and barometer. These sensors are generally integrated as an IMU.

- **Multi-mode Data Collection**: Data acquisition must account for various pedestrian movement patterns (e.g., normal walking, turning, standing) and different smartphone usage modes (e.g., handheld, swinging, in pocket, in bag), as PDR performance depends heavily on how the device is carried.

- **Magnetic Field Map Construction (Offline Phase)**: To enable infrastructure-free absolute positioning and initialization, magnetic field data (3-axis Earth's magnetic field values) must be collected extensively across different positions and floors in the target building to construct a magnetic field map (fingerprint map).

- **Ground Truth Generation**: Precise positional ground truth must be recorded simultaneously during walking experiments using highly accurate external systems (e.g., total station or LiDAR SLAM, although these are external to the final application, they are essential for validation) to accurately label the sensor data. Data sequences collected should correspond to paths or distances needed for testing localization accuracy.

2. **Data Preprocessing**: Raw sensor data from MEMS units is inherently noisy and highly sensitive to interference and usage orientation.

    - **Denoising and Filtering**: The collected raw inertial data (accelerometer, gyroscope, magnetometer) must be filtered to reduce noise, employing methods like Adaptive Gaussian Bilateral Filtering (AGBF).

    - **Data Alignment and Segmentation**: Sensor data streams are temporally aligned using timestamps. Data is then segmented using sliding windows to create samples for activity detection and Deep Learning (DL) training.

    - **Magnetometer Orientation Normalization**: Since magnetometer readings change based on the smartphone's arbitrary orientation, the magnetic field vectors must be rotationally transformed and aligned to a reference direction to ensure that the characteristics of a specific location can be accurately identified regardless of the device's orientation. This ensures robustness against varied phone poses.

3. **Feature Selection and Engineering**: Feature extraction focuses on converting raw IMU data into meaningful PDR components and localization characteristics.

    - **PDR Features**: Extract features required for Pedestrian Dead Reckoning, including:

        - **Step Detection**: Identification of foot/step events using methods like pitch amplitude or peak detection based on accelerometer data.

- **Step Length Estimation (SLE)**: Calculation of walking distance per step using linear, nonlinear, or adaptive models based on gait analysis.
- **Heading Estimation**: Calculation of movement direction, typically fusing gyroscope data with magnetometer data using sensor fusion techniques like the Linear Kalman Filter (LKF) or Extended Kalman Filter (EKF) to compensate for individual sensor errors.
- **3D Localization Features**: Barometer pressure readings are processed to determine altitude changes, enabling floor identification and 3D positioning.
- **Deep Learning Features**: Advanced temporal features (e.g., statistical features like maximum, minimum, percentiles) are extracted from the sensor data within the defined sliding windows for input into machine learning models.

4. **Model Selection**: The final localization solution requires a complex fusion model to combine the continuous PDR output with occasional absolute location corrections derived from the magnetic field map, thus mitigating the PDR's inherent drift error.

- **PDR Enhancement (Model Adaptation)**: Select a Deep Learning (DL) model, such as a Recurrent Neural Network (RNN) or Long Short-Term Memory (LSTM) network, to predict displacement (velocity and heading). This model is trained using data-driven PDR methods to accurately estimate movement across various smartphone usage modes.
- **3D Location Initialization Model**: A sequential DL model (e.g., LSTM) is used to analyze magnetic field sequences (e.g., from the user's first 10 steps) to detect the initial floor. Simultaneously, the initial 2D location on that floor is estimated by applying a matching algorithm, such as k-Nearest Neighbors (k-NN) regression, to compare the current magnetometer data against the magnetic field map.
- **Fusion and Correction Model:** To integrate the PDR output with the location corrections derived from the magnetic map, a robust filter-based fusion framework is necessary. This may involve:
  - **Particle Filter (PF):** The PF framework is highly suitable for non-linear indoor localization problems and for fusing IMU/PDR data with map constraints. PF models (potentially an Improved PF, or **IPF**) are used to combine magnetic positioning results with PDR.
  - **Map/Magnetic Constraints:** Magnetic field data is used to implement **Correction Nodes (CN)** in software or virtual landmarks to continuously correct the accumulated PDR drift errors without external infrastructure.
  - **Final Output:** The fused position is used to provide accurate location coordinates, which trigger the reminder functions defined by the application's requirements.

5. **Model Training:** The training process focuses on establishing robust localization models using data derived exclusively from the smartphone's built-in sensors, such as the Inertial Measurement Unit (IMU) and magnetometer.

   - **Data Collection and Fingerprinting (Offline Phase):**

     - **PDR Data Collection**: Collect time-series data using built-in **IMU sensors** (accelerometer, gyroscope, and magnetometer) to estimate position and direction. This data captures pedestrian movement across multiple **smartphone usage modes**, such as texting, swinging, pocket, and bag.

     - **Magnetic Fingerprint Construction**: To facilitate initial location estimation and drift correction, build a magnetic field map by collecting 3-axis magnetic field values at predetermined **Reference Points (RPs)** across the indoor environment, creating a unique signature for each location.

   - **PDR and Initialization:**

     - **PDR Deep Learning**: Train a Deep Learning (DL) model, such as an LSTM network, using the multi-modal sensor sequences to perform data-driven PDR. This model learns to predict pedestrian velocities and displacements, adapting to diverse smartphone holding modes.

     - **3D Initialization Model:** Train a sequential DL model (e.g., LSTM) using magnetic field sequences collected during walking (e.g., the first 10 steps) to predict the initial floor level. The initial horizontal 2D location estimation model uses the **k-Nearest Neighbors (k-NN) regression algorithm** trained on the static magnetic map.

   - **Fusion Framework Configuration:** The final estimation uses a filter-based fusion framework, such as the **Particle Filter (PF)**, or a Kalman Filter (KF/EKF/UKF), which must be configured with optimized parameters (e.g., noise covariance matrices) suitable for non-linear systems.

6. **Model Testing**: The testing phase is designed to evaluate the system's performance and robustness in real-world indoor environments, comparing predicted locations against established ground truth reference paths.

   - **Experimental Setup and Procedure**:

     - Testing must be carried out in complex **real-world indoor environments** (e.g., offices or multi-story buildings).

     - The system's performance is verified by having users traverse predetermined test paths, recording the application's predicted location against ground truth (actual path).

     - A critical aspect of evaluation is assessing robustness against real-world user variability, including different user activities and varying smartphone carrying poses (e.g., pocket, hand, swinging, texting).

7. **Feedback Loop**: Primary focus areas include mitigating the cumulative drift errors that plague PDR systems, often by strategically using magnetic field data as **Correction Nodes (CN)** or landmarks to reset the estimated trajectory.

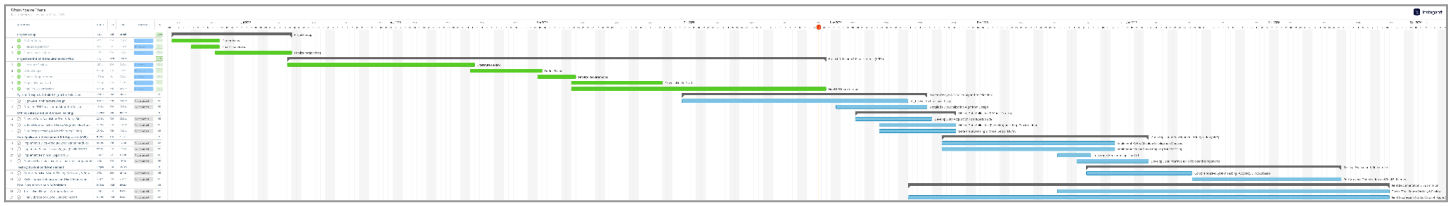
- **Drift and error correction optimization**:
  - Address the inherent limitation of PDR systems, which suffer from accumulated drift errors over time.
  - Refine the fusion weights or implementation of Correction Nodes (CN) based on magnetic field data. These CNs serve as software or virtual landmarks that periodically reset the PDR drift errors.
- **Enhancing generalization and pose invariance**:
  - Improve system robustness against varying user contexts (e.g., hand, pocket).
  - Refinement should focus on achieving **pose-invariant localization** by applying advanced sensor fusion techniques or orientation normalization methods to minimize performance degradation caused by changes in device placement.

3.4 Project Management Methodology

3.4.1 Schedule

3.4.1.1 Gantt Chart

Fig 1: Gantt chart



Due to difficulty visualizing the above gantt chart, the original could be found here.

3.4.1.2 Deliverables

All deliverables with their respective timelines could be viewed in the figure below.

Fig 2: Deliverables

- **Initial Project idea and supervisor selection** - 10th Jul 2025
- **Project Proposal Initial Draft** - 10th Oct 2025
- **Project proposal and Requirement Specification (15%)** - 13th Nov 2025
- **Prototype, Implementation, Interim Project Demonstration (15%)** - 2nd Feb 2026
- **Thesis Submission, Final prototype, Dataset, and Source Code submission (70%)** - 30th Mar 2026
- **Final Project Viva** - 15th Apr to 30th May 2026

3.4.2 Resource Requirements

The following highlights the data and skill requirements for successful completion of the project.

3.4.2.1 Data

- **Sensor Data for Pedestrian Dead Reckoning (PDR):** The primary raw data sources consist of continuous time-series measurements collected from the smartphone's built-in IMU sensors, specifically the accelerometer, gyroscope, and magnetometer. For multi-floor navigation capability, barometer sensor data must also be collected to detect pressure changes associated with floor transitions.

- **Diverse User Behavior Data:** Training data must capture variations in pedestrian activity, including basic walking patterns. Crucially, data must be collected across various smartphone usage poses (e.g., handheld, texting, swinging, pocket, bag) to enable the localization model to achieve pose-invariant localization that functions robustly despite device placement changes.

- **Localization Fingerprint Datasets (Magnetic Field):** A separate database must be constructed using magnetic field vectors collected at predefined Reference Points (RPs) across the indoor area to create a magnetic field map (fingerprint database). To minimize errors caused by the magnetometer's orientation dependency, these collected vectors must be rotationally transformed and aligned to a reference direction (e.g., map north) before constructing the map.

- **Ground Truth and Trajectory Annotation:** Accurate ground truth location coordinates must be obtained and timestamp-aligned with the collected sensor data sequences for training and rigorous testing. Due to inherent accumulation errors in ground truth collection methods, individual collected trajectories are typically constrained to limited distances (e.g., less than 200m).

- **Addressing Data Scarcity and Heterogeneity:** To enhance model generalization and prevent overfitting, especially when working with sequential magnetic field data, data augmentation techniques must be utilized (e.g., applying random rotation transformations or reversing magnetic sequences).

3.4.2.2 Skills

- **Algorithm optimization**: The ability to analyze and balance algorithm complexity with performance.
- **Problem-solving and adaptability**: Skill in designing rigorous testing procedures with the ability to efficiently minimize error, adapting to any unexpected outcome.
- **Technical documentation**: The ability to clearly document complex models and DL architectures required for the project.
- **Collaboration/Communication**: The ability to adapt one's depth of communication in order to communicate complex topics to a wider audience.

3.4.3 Risk and mitigation

Table 9: Risk and Mitigation

Risk	Sivererity	Frequency	Mitigation Strategy
PDR drift and cumulative error	5	5	Utilize filter-based sensor fusion frameworks like the Particle Filter (PF) or Kalman Filter (KF) to integrate continuous PDR output with absolute corrections. Implement Correction Nodes (CN) based on the magnetic field map, matching to periodically reset the PDR drift.
Localization performance is susceptible to magnetic interference	4	3	Apply adaptive constraints and outlier filtering algorithms to mitigate the impact of noisy magnetic data. Employ Deep Learning (DL) models trained on magnetic field sequences for robust heading estimation.
High computational cost and excessive battery drain	3	4	Sacrifice latency for more lightweight algorithms to minimize resource usage while satisfying real-time requirements.
Loss of vital documentation and files	5	4	Maintain a consistent backup routine for any and all documentation to cloud-based storage such as Google Drive or OneDrive.
Lack of domain and technical knowledge	5	4	Consult domain experts and expand the collection of relevant literature. Make use of online learning resources such as Stack Overflow, YouTube, etc.

3.5 Chapter Summary

This chapter outlines the development plan for a robust indoor localization system. It covers the selection of the Research Methodology (e.g., Positivism and Deductive approach) and the Development Methodology, which utilizes an iterative approach and Object-Oriented Analysis and Design (OOADM). Key technical sections detail methods to overcome single-sensor limitations by improving Pedestrian Dead Reckoning (PDR) and integrating absolute location data using magnetic constraints and core sensor fusion techniques like filtering (e.g., Particle Filters). The chapter also includes project management and risk mitigation strategies.

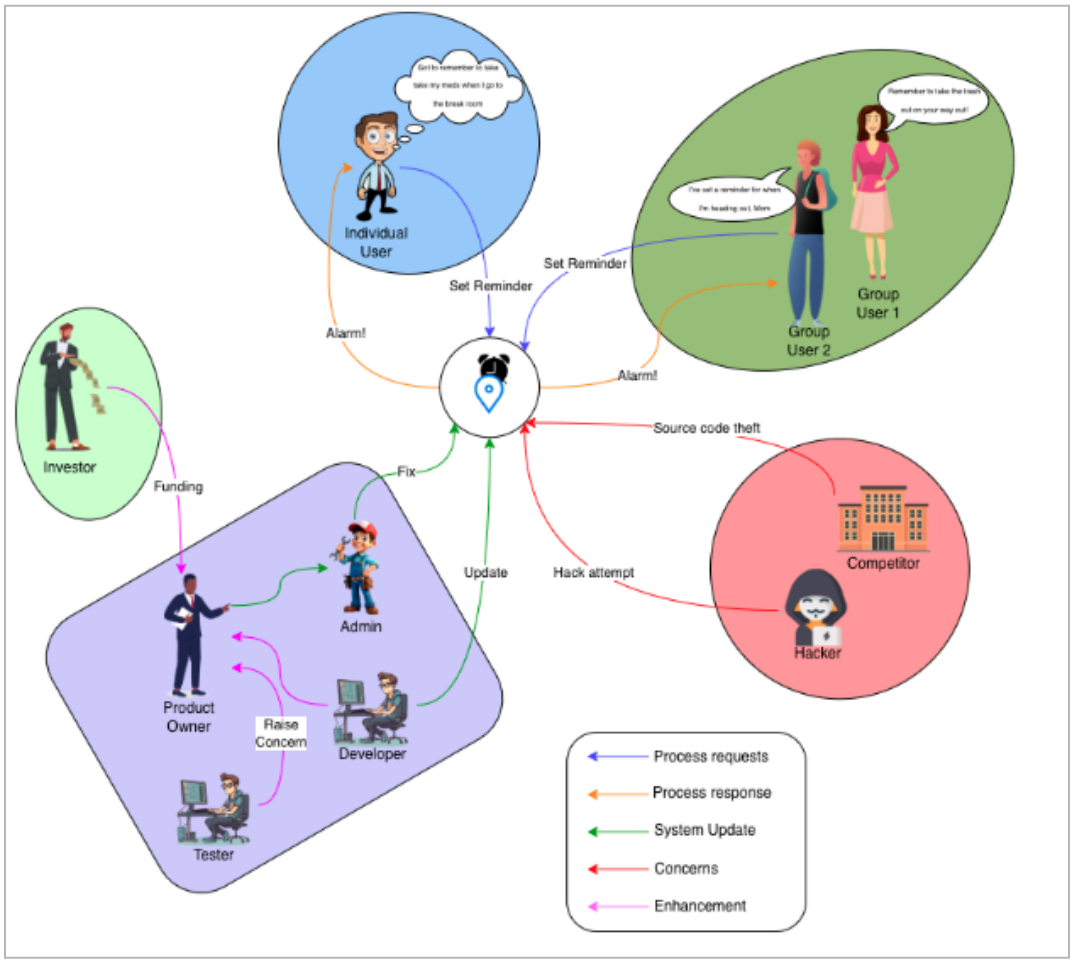
CHAPTER 04: SOFTWARE REQUIREMENT SPECIFICATION (SRS)

4.1 Chapter Overview

This chapter outlines the finalized system requirements, detailing the functional and non-functional requirements and the means by which they were each obtained. It also contains figures, tables, and diagrams to visualise various system interactions. Additionally, it uses the MoSCoW method to better communicate the priority assigned to each requirement.

4.2 Rich Picture Diagram

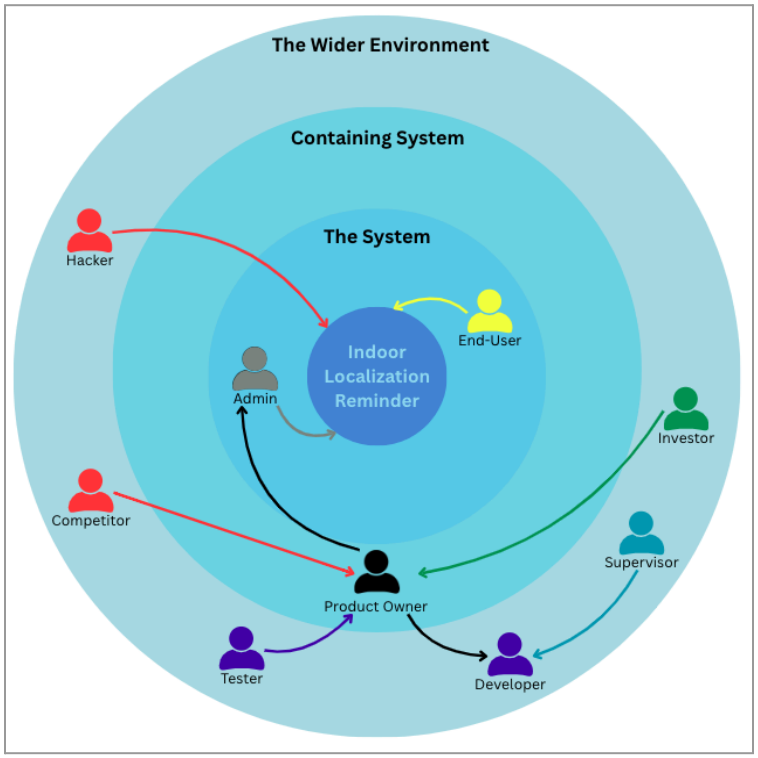
Fig 3: Rich Picture Diagram



4.3 Stakeholder Analysis

4.3.1 Stakeholder Onion Model

Fig 4: Onion Model



4.3.2 Stakeholder Viewpoints

Table 10: Stakeholder Viewpoints

Stakeholder	Role	Description
System Stakeholders		
Admin	System Maintenance	Address any issues with the system and ensure it is running smoothly.
End User	Application User	Use the application to set the location identities during initialization, followed by setting reminders for each when needed.
Containing System Stakeholders		
Product Owner	System Owner	Ensure that the system satisfies stakeholder demands while maintaining corporate objectives.
Wider Environment Stakeholders		

Hacker	Negative Stakeholder	Find weaknesses in the system to exploit and use data within the system maliciously.
Competitor	Negative Stakeholder	Provide similar services to users, thus dividing the user base and influencing market dynamics. Force innovation and constant improvement in the industry.
Tester	Quality Assurance	Test boundaries in an attempt to find and remediate any weaknesses in the system, thus improving the accuracy and security of the overall product.
Developer	System builder	Constantly improve the system to enhance accuracy and user experience to meet the requirements set by the product owner.
Supervisor	Advisor	Guide the project team, ensuring the efficient and timely completion of milestones.
Investor	Financial Beneficiary	Provide financial aid for project development with the expectation of return on investment based on product performance in the market.

4.4 Selection of Requirement Elicitation Methodologies

For this project, a few different forms of methods were used to gather requirements. Each one was chosen based on its ability to gather detailed insight accurately for relevant aspects of the project development.

Table 11: Requirement Elicitation Methodologies

Method 1: Literature Review
A deep investigation was conducted into the literature in the field of indoor localization to find information on existing systems, along with their respective gaps, and discussions on future points of improvement for each, with the added benefit of gathering useful methods and best practices to carry out this project.
Method 2: Surveys
This was done to gather first-hand opinions of possible end users to identify end-user requirements, ensuring that the final product completes its intended purpose while maintaining user satisfaction by fitting into their expectations.
Method 3: Brainstorming

Brainstorming sessions were conducted to refine system features and generate ideas for system features and functionalities. These sessions, typically held with key stakeholders or individuals who have implemented similar systems, were intended to gather relevant, accurate, and actionable insights.

4.5 Discussion of Findings

For each method of requirement elicitation used, its relevant findings are listed below.

4.5.1 Literature Review Findings

Table 12: Literature Review Findings

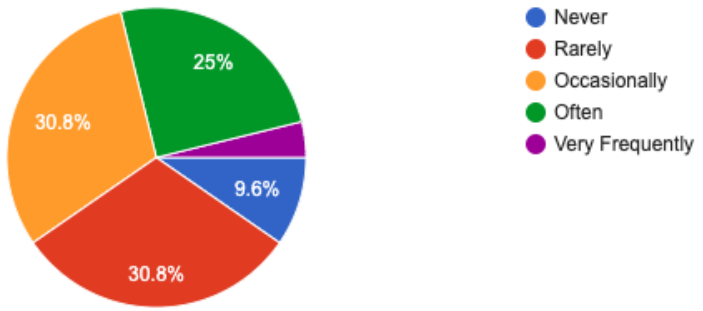
Citation	Finding
(Liu et al., 2025)	No single indoor localization technology accessible via standard smartphone sensors offers a universal solution regarding accuracy, cost, power consumption, and portability. Therefore, fusing multi-source sensors is essential, benefiting from the complementary advantages of various techniques to achieve pervasive and reliable indoor localization.
(Poulose et al., 2019)	Pedestrian Dead Reckoning (PDR), which utilizes the built-in Inertial Measurement Unit (IMU) sensors, suffers from accumulating drift errors as the pedestrian walking distance increases, consequently reducing location accuracy due to inherent smartphone sensor errors.
(Kim & Shin, 2025)	A major challenge for PDR systems is adapting to the unconstrained use of smartphones; localization performance degrades significantly because the algorithms must maintain pose-invariant performance across various device-carrying positions and orientations (e.g., in the hand, pocket, or backpack).
(Zhou et al., 2024)	Deep learning (DL) architectures are increasingly utilized in PDR systems to mitigate accumulated errors and account for the effects of varied smartphone usage modes by producing more accurate predictions of user displacements than conventional PDR methods.
(Qi et al., 2024)	The success of fusion strategies depends on the architecture being used. Tightly coupled structures are advantageous because they utilize the raw observations from each positioning source directly as measurement inputs for the integrated filtering algorithms.
(Wang et al., 2025)	Traditional filtering algorithms often employed for sensor fusion depend heavily on making strict assumptions regarding system models and noise distributions, which limits their effectiveness when deployed in the dynamic and uncertain conditions of real-world environments.

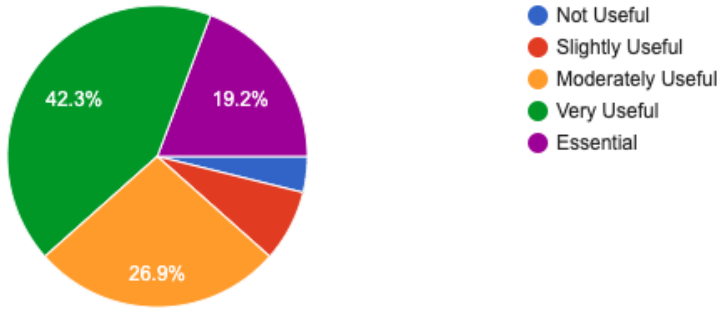
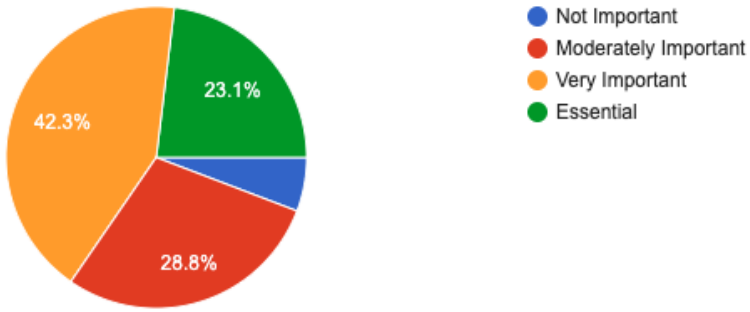
4.5.2 Interview Findings

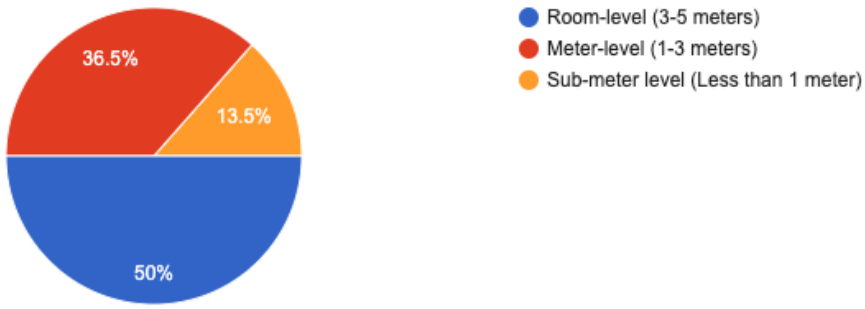
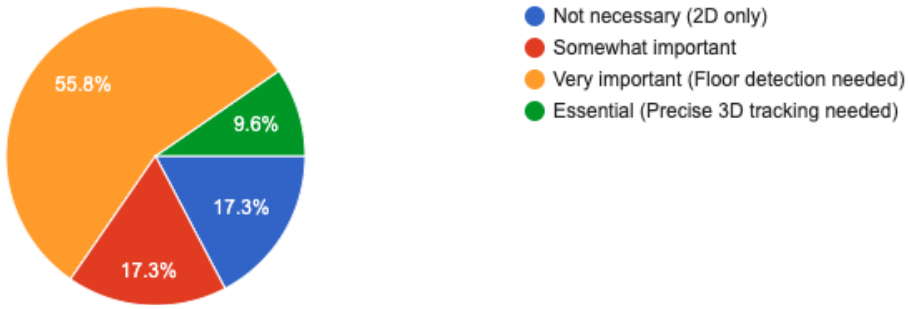
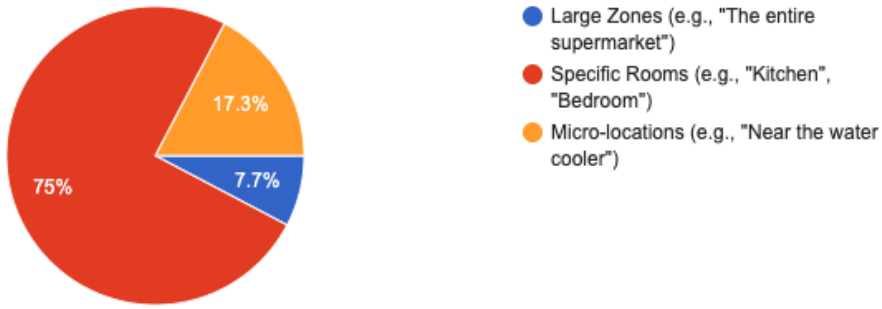
Due to issues with time-of-availability mismatches, the author has postponed the conduct of interviews, as the overall credibility of the findings is not adversely affected since the extensive archival research, deployment of a survey, and conduct of brainstorming sessions have all established the foundational technical requirements, challenges, and limitations of the project being proposed.

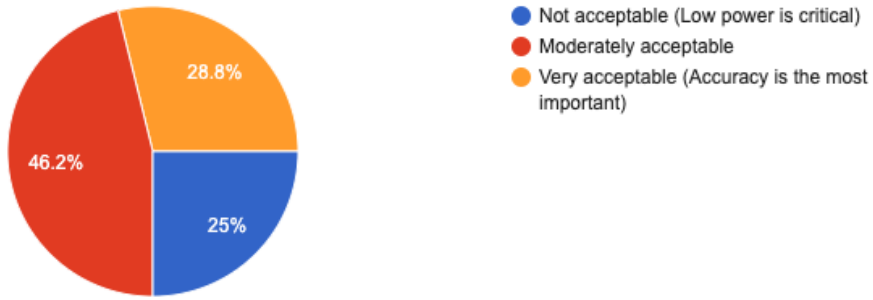
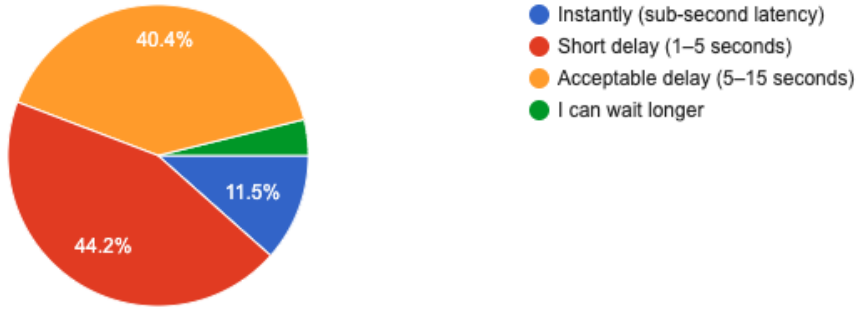
4.5.3 Survey Findings

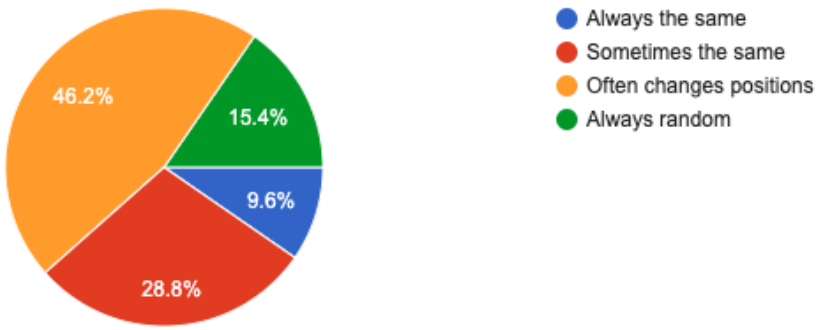
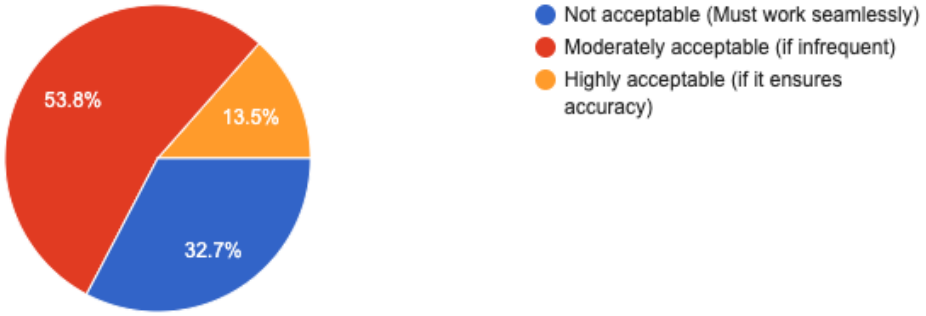
Table 13: Survey Findings

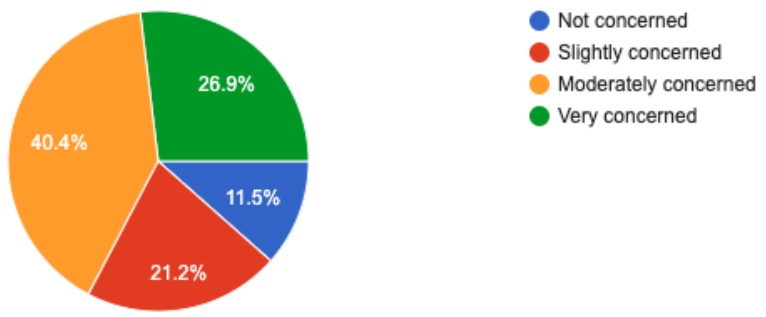
How frequently do you rely on location or navigation services when inside large indoor spaces (e.g., hospitals, large office buildings, malls)?	
Aim	To gauge the general demand for indoor localization services, showing that LBS is increasingly needed indoors.
Results	
Finding & Conclusion	30.8% of participants responded as both “Rarely” and “Occationally”. This either shows that there is low demand or low availability of similar solutions, which means the unawareness of such a solution is responsible for these results.
How useful would an application be that automatically triggers a reminder or alert when you reach a specific place indoors (e.g., "Take your medicine" when you enter a specific room)?	
Aim	To validate the specific need for a reminder application linked to indoor location (a user experience-centric application).

Results	
Finding & Conclusion	42.3% respondents responded that such a solution would be “Very Useful” while 19.2% stated that it would be essential. An overall 61.5% validated the need for such a solution, which proves the latter conclusion for the previous question.
If such an application were available, how important is it that it functions reliably using ONLY your smartphone's built-in sensors (i.e., no external Wi-Fi infrastructure or dedicated hardware needed)?	
Aim	To validate the need for an infrastructure-free approach, which justifies the use of PDR/IMU/Barometer technology, despite its inherent complexity.
Results	
Finding & Conclusion	A combined total of 65.1% support the notion that such a solution would be more effective if it were to be infrastructure-independent, running solely using smartphone components.
For a reminder to be considered useful, what is the maximum acceptable location error (accuracy) for the system?	
Aim	To identify the target localization accuracy. Achieving meter-level accuracy (or better) is a goal for high-precision systems.

Results	
Finding & Conclusion	50% accept room-level error to be acceptable, which allows for a slight reduction in accuracy to balance out the overall efficiency, as room-level error is quite large.
How crucial is it that the application can accurately track your location across multiple floors in a building?	
Aim	To determine if the system requires 3D localization, which necessitates the use of the barometer sensor for floor identification.
Results	
Finding & Conclusion	55.8% and another 9.6% support the implementation of floor-level detection, indicating a need for the use of the barometer sensor for floor identification.
Which types of locations would you most likely set reminders for?	
Aim	To understand the granularity of indoor features needed for location matching (e.g., structural landmarks like corners or specific zones).
Results	 17.3% 75% 7.7% Large Zones (e.g., "The entire supermarket") Specific Rooms (e.g., "Kitchen", "Bedroom") Micro-locations (e.g., "Near the water cooler")

Finding & Conclusion	75% of the respondents would use the application on a room-identification level, which indicates that absolute accuracy could be reduced to increase the priority of efficiency.
Since continuous tracking uses phone sensors, how acceptable is increased battery drain (energy consumption) for improved localization accuracy?	
Aim	To gauge user tolerance for high power consumption, a major challenge in IMU/PDR systems.
Results	
Finding & Conclusion	46.2% of respondents were moderately accepting, with a further 28.8% being completely accepting of increased battery usage. This proves that an overall increase in power consumption is acceptable, but overall power maintenance should be considered.
How quickly must the application register your arrival and trigger the reminder once you have reached the target location?	
Aim	To define the required speed of response, which impacts the computational complexity of the real-time processing algorithms (e.g., filters).
Results	
Finding & Conclusion	An overall majority of 84.6% accept an average delay of 1-15 seconds. This means that some latency is acceptable and that a maximum of 15 seconds would be tolerated by end-users.

How often does the position you carry your smartphone change while walking indoors (e.g., handheld, in pocket, in bag)?	
Aim	To assess the complexity needed for pose-invariant localization, as PDR accuracy is heavily influenced by how the device is carried.
Results	
Finding & Conclusion	A total of 75% respondents confirmed the average change in phone carrying position, indicating that pose-invariant performance is necessary.
If the application required a short, specific action (like standing still or shaking the phone) once per hour to correct accumulated location drift errors, how acceptable would this be?	
Aim	To determine user tolerance for calibration interruptions necessary to mitigate drift errors inherent in IMU/PDR tracking.
Results	
Finding & Conclusion	53.8% of respondents were moderately accepting of calibration interruptions, allowing for the consideration of the possibility of minor user-performed calibration steps to improve localization accuracy.
How concerned are you about the privacy and security of location data that tracks your movements within private indoor spaces?	
Aim	Location privacy is a critical concern, especially when using mobile devices for tracking. This helps define security and data handling priorities.

Results	
Finding & Conclusion	A majority of respondents showed concern for the collection of location data, indicating that a non-intrusive data collection method is necessary.

4.5.4 Brainstorming

Table 14: Brainstorming Findings

Criteria	Finding
Project relevance and significance	High demand exists for infrastructure-independent localization because people spend most of their time indoors, where traditional GNSS signals fail. The project addresses the need for user experience-centric applications like location-based reminders.
Novelty and innovation	The use of Deep Learning (DL)-based Pedestrian Dead Reckoning (PDR) is necessary for improved accuracy and stability when pedestrians use their smartphones in multiple usage poses. A key innovation is using a magnetic field map combined with DL (LSTM) to perform accurate 3D location initialization based on the user's first few steps, eliminating the need for a known starting point.
Technical Feasibility	The project is feasible because modern smartphones contain the necessary Micro-Electro-Mechanical System (MEMS) sensors to implement PDR and floor detection. While continuous tracking systems using IMU/PDR face challenges with high power consumption, PDR algorithms typically have a low computational load, making real-time processing achievable.
Addressing limitations of existing methods	The proposed approach mitigates the critical issues of conventional PDR systems like cumulative error, start location, and pose dependency.
Data considerations	The system relies entirely on raw sensor data collected from the built-in IMU sensors and the barometer. This data must be preprocessed and analyzed using specialized methods like Deep Learning networks for step detection, velocity/heading estimation, and activity classification. An offline magnetic field map must be constructed to serve as the reference database for initial location estimation and drift correction nodes.

Technical contributions and documentation	The project will contribute a robust fusion methodology specifically designed for infrastructure-free multi-floor indoor localization, coupled with thorough documentation for future enhancement and development of indoor localization.

4.6 Summary of Findings

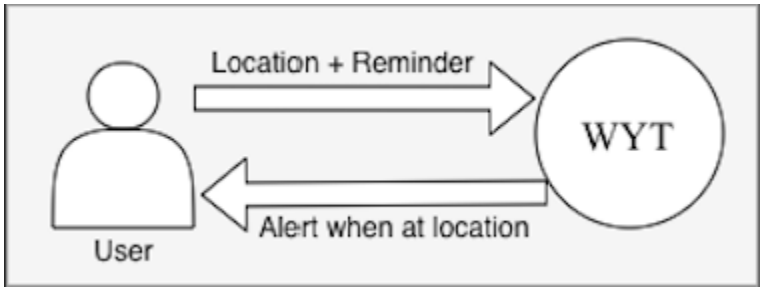
Table 15: Summary of Findings

Finding	Literature review	Survey	Brainstorming
No single indoor localization technology offers a universal solution; therefore, fusing multi-source sensors is essential to achieve reliable indoor localization.	✔		
Pedestrian Dead Reckoning (PDR) systems suffer from accumulating drift errors as walking distance increases, necessitating mitigation strategies.	✔		✔
Localization performance degrades significantly due to the difficulty of maintaining pose-invariant performance across various device-carrying positions, as most people have constantly changing device-carrying positions.	✔	✔	
Deep learning (DL) architectures are increasingly utilized in PDR systems to mitigate accumulated errors and account for the effects of varied smartphone usage modes.	✔		✔
Tightly coupled fusion architectures are advantageous as they utilize raw observations directly as measurement inputs for integrated filtering algorithms.	✔		
Traditional filtering algorithms depend heavily on strict assumptions regarding system models and noise distributions, limiting their effectiveness in dynamic real-world environments.	✔		
High demand exists for infrastructure-independent localization because people spend most of their time indoors.		✔	✔
There is strong support for an infrastructure-independent solution that runs solely using the smartphone's built-in components.		✔	✔
A low user reliance/low perceived demand suggests low availability or unawareness of existing solutions.		✔	

Room-level error is acceptable, allowing for a slight reduction in absolute accuracy to balance overall efficiency.		✔	
There is a need for 3D/floor-level detection, indicating the necessity of using the barometer sensor for floor identification.		✔	
An overall increase in power consumption is acceptable for improved localization accuracy, but overall power maintenance should still be considered.		✔	
Some latency is acceptable, with an average delay of 1–15 seconds being tolerated by end-users.		✔	
A key innovation involves using a magnetic field map combined with DL (LSTM) for accurate 3D location initialization based on the user's first few steps, eliminating the need for a known starting point.			✔
The project is technically feasible because modern smartphones contain the necessary MEMS sensors for PDR and floor detection, and PDR algorithms have a low computational load.			✔
The proposed approach must mitigate critical PDR issues such as cumulative error, unknown start location, and pose dependency.	✔		✔
The system relies entirely on raw sensor data analyzed by specialized Deep Learning networks, and an offline magnetic field map must be constructed for reference.			✔

4.7 Context Diagram

Fig 5: Context Diagram



4.8 Use Case Diagram

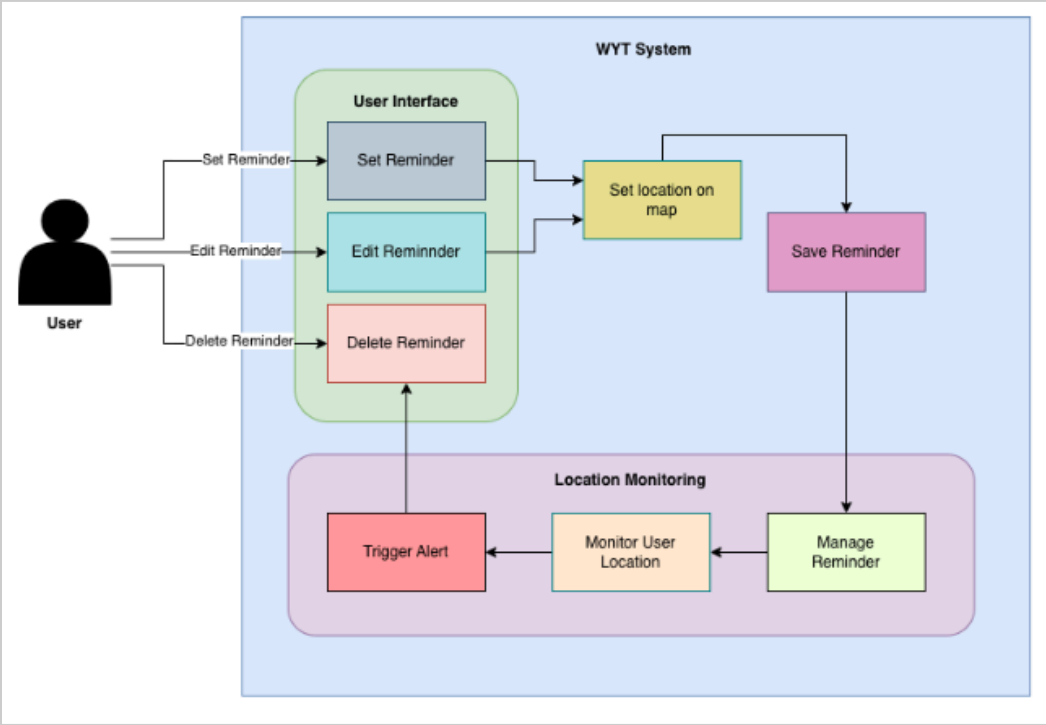


Fig 6: Use Case Diagram

4.9 Use Case Description

Table 16: Use Case Description-1

Use Case	WYT System
Name	Set Reminder
Description	A reminder is set for a specific location trigger
Primary Actor	User
Supporting Actor	None
Preconditions	The location must be predefined as a room/place of significance during the initial setup of the application.
Extended Use Cases	Edit Reminder, Delete Reminder
Included Use Cases	None
Alternative Flow	None
Exceptional Flow	If setting the reminder fails, the system logs the error and provides a description of the cause to the user, prompting the user to retry.

Post Condition	The reminder is set, and the proximity to the trigger location is constantly monitored by the system till it is close enough to trigger the alert.
Comment	This is the main function of the application.

Table 17: Use Case Description-2

Use Case	WYT System
Name	Delete Reminder
Description	An existing reminder is deleted.
Primary Actor	User
Supporting Actor	None
Preconditions	A reminder must already be set for the location specified.
Extended Use Cases	None
Included Use Cases	Set Reminder
Alternative Flow	None
Exceptional Flow	If deleting the reminder fails, the system logs the error and provides a description of the cause to the user, prompting the user to retry.
Post Condition	The reminder is deleted successfully.
Comment	This is dependent on the existence of a specified reminder.

4.10 Requirements

The MoSCoW principle is used for prioritization of system requirements as follows:

- **M:** Must have (mandatory)
- **S:** Should Have (important)
- **C:** Could Have (desirable)
- **W:** Will Not Have (out of scope)

4.10.1 Functional Requirements

Table 18: Functional Requirements

ID	Requirement	Priority
FR01	The system must execute Pedestrian Dead Reckoning (PDR) to estimate the user's continuous relative displacement (velocity, distance, and heading) using only smartphone inertial sensors.	M
FR02	The system must process PDR data using Deep Learning (DL) or data-driven techniques to ensure accurate positioning across multiple smartphone usage modes (e.g., pocket, swinging, texting).	S
FR03	The system must detect and identify floor transitions (inter-floor movements) reliably using the smartphone's built-in barometer sensor.	M
FR04	The system must implement an autonomous drift correction mechanism (e.g., using magnetic field map matching or semantic map cues) to reset accumulated PDR error.	S
FR05	The application must allow the user to set a location-based reminder and trigger a notification/alert when the user reaches a predefined 3D coordinate or zone.	M
FR06	The system should perform 3D location initialization (initial floor and 2D position) automatically upon startup, potentially using magnetic field sequence analysis, without requiring the user to specify a known starting location.	S
FR07	The system could provide a user interface for viewing the estimated trajectory overlaid on a stored indoor topological map or floor plan.	W

4.10 Non-Functional Requirements

Table 19: Non-Functional Requirements

ID	Non-Functional Requirement	Description	Priority
NFR 01	Accuracy	The localization system must achieve meter-level accuracy (e.g., ≤1.0 meter mean error) to reliably distinguish room-level locations for reminders	M
NFR 02	Infrastructure Independence	The system must function entirely using the smartphone's built-in sensors and an offline map/database, requiring no external fixed infrastructure.	M
NFR 03	Robustness	The system must provide stable and continuous localization across multi-floor environments and diverse pedestrian movements (e.g., standing, walking, turning).	S

NFR 04	Real-time performance	The application must process sensor data and trigger reminders with low latency, satisfying real-time application requirements (e.g., updating location within 1−15 seconds of movement)	M
NFR 05	Energy efficiency	The system should minimize computational complexity and limit battery drain associated with continuous sensor tracking and processing	S
NFR 06	Scalability	The PDR algorithms should be generalized and maintain performance when deployed in new, unseen indoor environments without extensive retraining	C
NFR 07	Privacy	The processing of sensitive location data must occur entirely on the user's local smartphone device to ensure location privacy and security	M

4.11 Chapter Summary

The Software Requirement Specification (SRS) chapter provided a comprehensive overview of the system's requirements, beginning with the visualization of the system's context and its interactions with various stakeholders using tools such as the Rich Picture Diagram and the Stakeholder Onion Model. This process included documenting stakeholder viewpoints and incorporating findings from various elicitation methodologies (such as literature review, surveys, and brainstorming) to determine critical system features. The chapter culminated in the formal definition of the system’s Functional and Non-Functional Requirements, which were rigorously prioritized using the MoSCoW principle to focus development efforts on features deemed mandatory for the system's success.

CHAPTER 05: SOCIAL, LEGAL, ETHICAL AND PROFESSIONAL ISSUES (SLEP)

5.1 Chapter Overview

This chapter establishes the governing ethical standards and professional obligations for members of the British Computer Society (BCS), setting rigorous rules that often exceed general legal requirements to ensure integrity within the IT profession. It delineates the scope of member responsibilities toward the public, employers, and the industry at large, while establishing a framework for disciplinary action should these professional standards be breached.

5.2 SLEP issues and mitigation

Social ● **Issue (Inclusivity):** The project is designed to assist individuals with memory impairments (e.g., Alzheimer's, dementia, ADHD) and general pedestrians. A social risk exists if the system is inaccessible, potentially causing distress or missed critical tasks (e.g., medication reminders) for vulnerable users. ● **Mitigation:** The application is designed to run on standard commercial smartphones without requiring expensive external hardware (like UWB or beacons), promoting inclusivity and equal access for users who cannot afford proprietary systems.	**Legal** ● **Issue (Data Privacy Legislation):** The system relies on continuously tracking the user's indoor location. Collecting and storing movement data raises significant legal issues regarding compliance with data protection laws (e.g., GDPR), as location data is considered sensitive personally identifiable information (PII). ● **Mitigation:** To comply with privacy legislation and protect user rights, the project mandates that **all processing of sensitive location data must occur entirely on the user's local smartphone device** (Edge Computing). This "In-Scope" privacy requirement ensures that raw sensor data is not transmitted to external cloud servers, minimizing the risk of unlawful interception or data breaches.
Ethical ● **Issue (Surveillance & Consent):** There is an ethical risk regarding the "tracking" nature of the application. Even if data is local, the software functions as a surveillance tool. Ethical concerns arise if the user is unaware of the extent of the tracking (e.g., recording frequency or history). ● **Mitigation:** The system is designed as a user-centric tool where the user explicitly defines trigger areas. The project addresses this by ensuring transparency in how data is	**Professional** ● **Issue (Technical Complexity & Integrity):** The project involves implementing complex Deep Learning models (xLSTM) and advanced sensor fusion algorithms (Particle Filters). Claiming high accuracy without rigorous validation or possessing the necessary skills would breach professional integrity. ● **Mitigation:** The developer acknowledges the need for specific skills and mitigates the competence risk by consulting domain experts

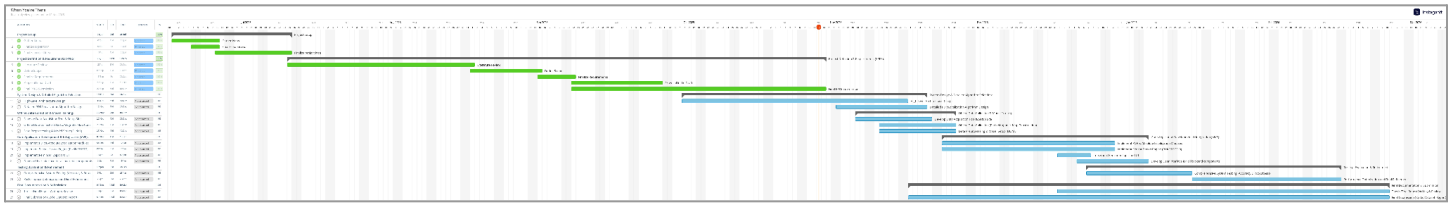
used—specifically for real-time positioning updates rather than historical surveillance. The use of widely available datasets (UJIIndoorLoc) for training rather than covertly collecting data from unknowing participants during the development phase adheres to ethical research standards.	and utilizing online learning resources. Furthermore, the project includes a rigorous **Testing and Evaluation** phase using standard metrics to scientifically validate performance claims against ground truth data, ensuring the professional delivery of a functioning system rather than making unsubstantiated claims.

5.3 Chapter Summary

The Code of Conduct articulates four essential tenets—Public Interest, Professional Competence and Integrity, Duty to Relevant Authority, and Duty to the Profession—that collectively define the ethical boundaries for IT practitioners. These principles, supported by specific interpretative guidance, mandate that members prioritize public wellbeing and privacy, operate strictly within their areas of competence, avoid conflicts of interest, and actively uphold the reputation of the information technology field.

CHAPTER 08: TIME SCHEDULE

Fig 7: Time Schedule Gantt



Due to difficulty visualizing the above gantt chart, the original could be found here.

The Gantt chart above is the same as found in chapter 3 as both copies are up to date up until the time of this document completion.

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APPENDIX A: USE CASE DESCRIPTION-3

Table 20: Use Case Description-3

Use Case	WYT System
Name	Edit Reminder
Description	An existing reminder is edited such that the description or trigger location is changed.
Primary Actor	User
Supporting Actor	None
Preconditions	A reminder must already be set for the location specified.
Extended Use Cases	None
Included Use Cases	Set Reminder
Alternative Flow	None
Exceptional Flow	If editing the reminder fails, the system logs the error and provides a description of the cause to the user, prompting the user to retry.
Post Condition	The reminder is edited successfully.
Comment	This is dependent on the existence of a specified reminder.